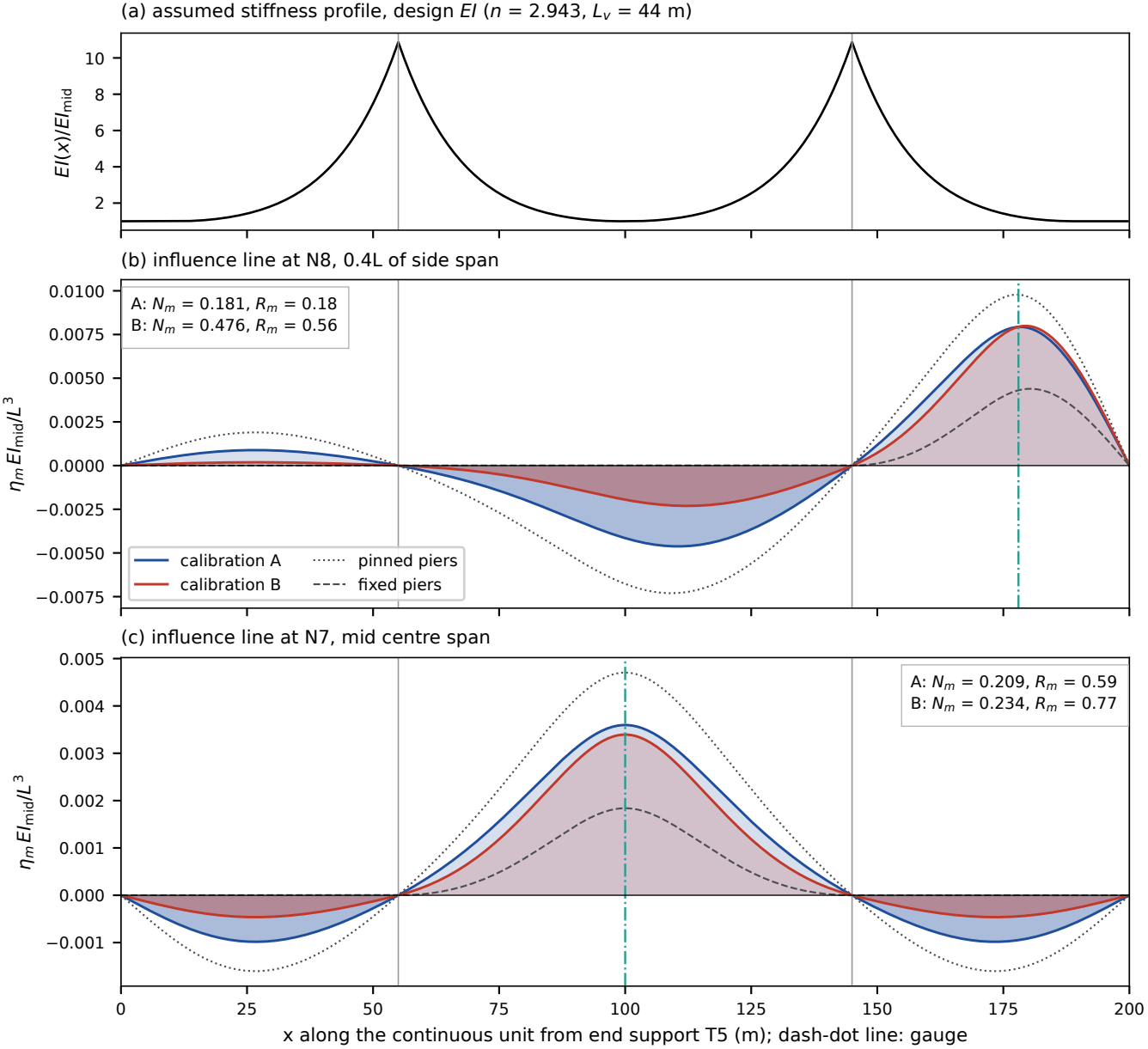

(a) assumed stiffness profile, design EI (n = 2.943, L_v = 44 m)
EI(x)/EI_mid
10
8
6
4
2
(b) influence line at N8, 0.4L of side span
A: N_m = 0.181, R_m = 0.18
B: N_m = 0.476, R_m = 0.56
η_m EI_mid/L^3
0.0100
0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
calibration A
calibration B
pinned piers
fixed piers
(c) influence line at N7, mid centre span
A: N_m = 0.209, R_m = 0.59
B: N_m = 0.234, R_m = 0.77
η_m EI_mid/L^3
0.005
0.004
0.003
0.002
0.001
0.000
−0.001
0
25
50
75
100
125
150
175
200
x along the continuous unit from end support T5 (m); dash-dot line: gauge